Quasi-likelihoods

Statistics III - CdL SSE

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- In Unit C and Unit D we saw that binomial and Poisson regression may suffer from **overdispersion**: the data are more variable than the model allows.
- This unit presents **quasi-likelihoods** (Wedderburn 1974), a simple way to handle overdispersion **without** specifying a distribution for Y_i: we only specify its **mean** and its **variance**.
- The unit is organized in three parts:
  - a **motivating example**, the `Ants` data;
  - the **theory** of quasi-likelihood models;
  - the **complete analysis** of the example.
- An alternative, **not covered** in this course, is to replace the Poisson or the binomial with a different parametric distribution, e.g. the negative binomial.

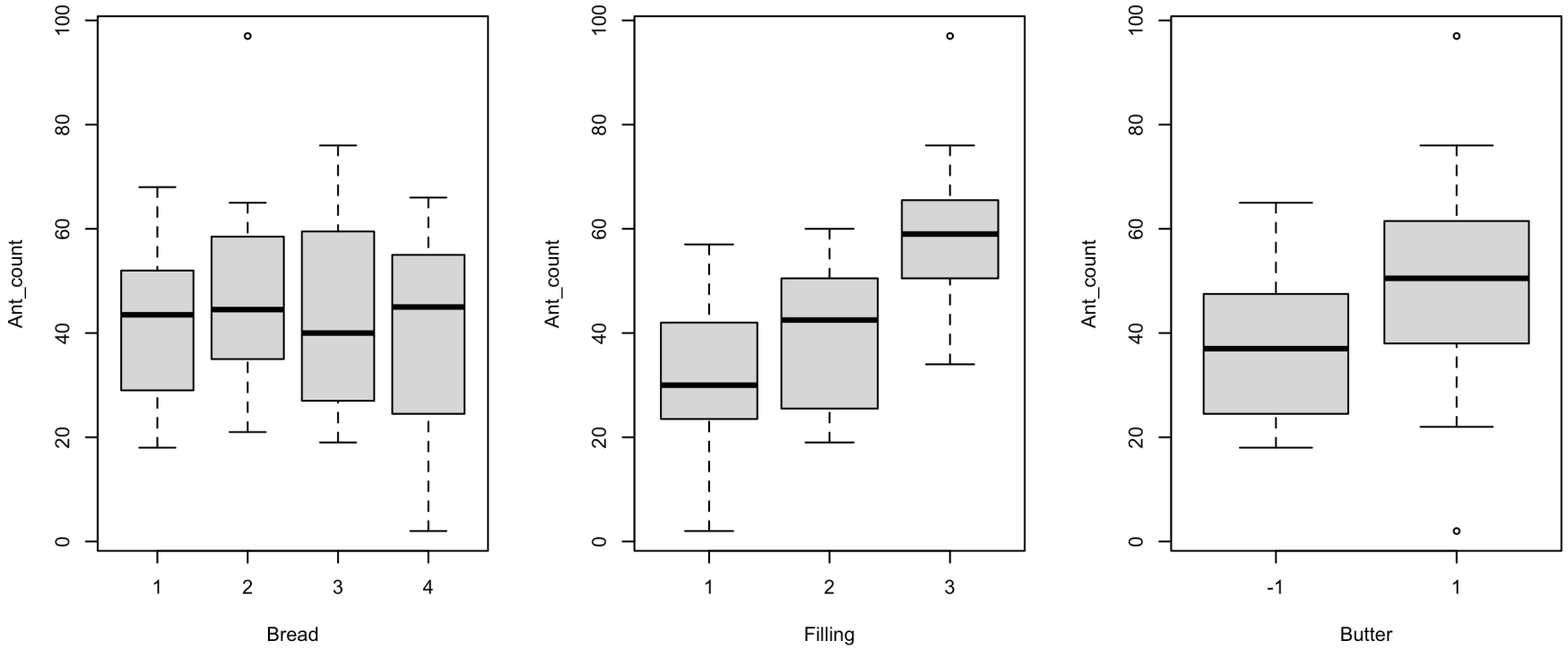
> The content of this Unit is covered in **Chapter 6** of Salvan et al. (2020). Alternatively, see **Sections 8.1 and 8.3** of Agresti (2015).

A motivating example

The `Ants` data

- The `Ants` dataset of the `MLGdata` library comes from an experiment run by students of an Australian university (Salvan et al. 2020). The aim is to study which sandwiches **meat ants** prefer.
- The **explanatory variables** are three experimental factors:
  - `Bread` (4 types): rye, wholemeal, multigrain, white;
  - `Filling` (3 types): Vegemite, peanut butter, ham and pickles;
  - `Butter` (2 levels): present (1) or absent (-1).
- For each of the $4 \times 3 \times 2 = 24$ combinations, **two** pieces of sandwich were placed near the entrance of an ant nest, for a total of $n = 48$ observations.
- The **response** `Ant_count` is the number of ants on the sandwich after 5 minutes.

Exploratory analysis



- `Filling` and `Butter` seem to matter, whereas `Bread` has a weaker effect.
- The response is a **count**, therefore a natural starting point is a **Poisson regression** model with logarithmic link, including the `Filling`-`Butter` interaction ($p = 9$).

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A Poisson regression model

	Estimate	Std. Error	p-value
(Intercept)	3.386	0.075	$<$0.001
Bread2	0.154	0.061	0.011
Bread3	0.026	0.063	0.684
Bread4	-0.030	0.064	0.634
Filling2	0.097	0.088	0.272
Filling3	0.468	0.081	$<$0.001
Butter1	0.063	0.089	0.478
Filling2:Butter1	0.248	0.119	0.038
Filling3:Butter1	0.277	0.111	0.012

- According to the Wald tests, `Bread2`, `Filling3` and both **interaction** terms are significant at the 5% level.
- But can we trust these standard errors?

Is there overdispersion?

- The residual deviance is $D(\boldsymbol{y}; \hat{\boldsymbol{\mu}}) = 183.05$ and the **Pearson statistic** of Unit B is $X^2 = 163.07$, both on $n - p = 39$ degrees of freedom. The p-value of the X^2 test is <0.001.
- Both statistics are **more than four times** the degrees of freedom: the data are **much more variable** than implied by the Poisson model, in which $\mathrm{var}(Y_i) = \mu_i$.
- The Poisson model is **misspecified** and its standard errors are **too small**: the significant coefficients of the previous slide should not be trusted.

A large X^2 may also be caused by a **misspecified mean**, e.g. an omitted covariate, a missing interaction or a wrong link function. Correcting the standard errors is appropriate **only if** the mean structure $g(\mu_i) = \boldsymbol{x}_i^T \beta$ is correct.

A word of caution

- A tempting shortcut is to compare the sample mean and variance of the response. In the `Ants` data they are 43.6 and 327. Is this enough to conclude that there is overdispersion? **No**.
- The Poisson assumption concerns Y_i **given the covariates**. Consider $n = 2000$ observations from

$$Y_i \overset{\text{ind}}{\sim} \text{Poisson}(\mu_i), \qquad \log \mu_i = 1 + 3x_i, \qquad x_i \overset{\text{iid}}{\sim} \text{Bernoulli}(0.5),$$

  which have **no overdispersion** by construction.
- The simulated data have mean 28.2 and variance 691.2, simply because the covariate shifts the mean. Once the covariate is accounted for, the Poisson fit gives $X^2/(n-p) = 1.009$.

> Back to the `Ants` data: how can we fix the standard errors, **without** having to choose a new distribution for the counts?

Quasi-likelihoods

A motivating remark: the linear model

- In the **Gaussian linear model** $\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{\epsilon}$ with $\epsilon_i \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma^2)$, the maximum likelihood estimator $\hat{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$ has the **exact** distribution

$$\hat{\beta} \sim \mathrm{N}_p(\beta, \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}).$$

- In Unit A we saw that under the much weaker **second-order assumptions** $\mathbb{E}(\boldsymbol{\epsilon}) = \mathbf{0}$ and $\mathrm{var}(\boldsymbol{\epsilon}) = \sigma^2 I_n$, the same estimator is still **unbiased**, with $\mathrm{var}(\hat{\beta}) = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$, and **asymptotically normal**.

> If the normal model is misspecified but the second-order assumptions hold, the inference based on the normal model remains **valid for large** n. Gaussianity is only needed for **exact** results.

- **Question**: does the same happen for the maximum likelihood estimator of a **GLM**?

Likelihood equations of a GLM

- Let $Y_i \overset{\text{ind}}{\sim} \text{ED}(\mu_i, a_i(\phi)v(\mu_i))$ with $g(\mu_i) = \boldsymbol{x}_i^T\beta$ and $a_i(\phi) = \phi/\omega_i$. Recall from Unit B that the **likelihood equations** are

$$\sum_{i=1}^{n} \frac{(y_i - \mu_i)}{\text{var}(Y_i)} \frac{\partial \mu_i}{\partial \beta_r} = \frac{1}{\phi} \sum_{i=1}^{n} \omega_i \frac{(y_i - \mu_i)}{v(\mu_i)} \frac{x_{ir}}{g'(\mu_i)} = 0, \qquad r = 1, \ldots, p.$$

- These equations depend on the distribution of Y_i **only** through

$$\mathbb{E}(Y_i) = \mu_i = g^{-1}(\boldsymbol{x}_i^T\beta), \qquad \text{var}(Y_i) = \frac{\phi}{\omega_i} v(\mu_i).$$

- Higher-order moments, such as the skewness, have **no impact** on the equations: any distribution with the same first two moments leads to the **same** estimate $\hat{\beta}$. They only affect the **distribution** of $\hat{\beta}$, e.g. how quickly it approaches normality.

The quasi-likelihood model

The **quasi-likelihood model** only assumes the following **second-order** conditions:

- **(E.1)** $\mathbb{E}(Y_i) = \mu_i = g^{-1}(\boldsymbol{x}_i^T \beta)$;
- **(E.2)** $\text{var}(Y_i) = \dfrac{\phi}{\omega_i} v(\mu_i)$, with $\phi > 0$ **unknown**;
- **(E.3)** Y_i and Y_j are independent for $i \neq j$.

- This is a **semiparametric** model: only the first two moments of Y_i are specified.
- The estimate $\hat{\beta}$ is defined as the solution of the **estimating equations**

$$\sum_{i=1}^{n} \omega_i \frac{(y_i - \mu_i)}{v(\mu_i)} \frac{x_{ir}}{g'(\mu_i)} = 0, \qquad r = 1, \ldots, p,$$

which do not depend on ϕ and can be solved with the **IRLS** algorithm of Unit B, **whatever** the choice of $v(\cdot)$.

Variance inflation

- For the Gaussian model, **(E.2)** is the usual homoscedasticity assumption, with $\phi = \sigma^2$. The novelty concerns **counts** and **proportions**, for which the GLM imposes $\phi = 1$.
- Keeping the variance function of a GLM, **(E.2)** **inflates** its variance by the factor ϕ. We have **overdispersion** if $\phi > 1$ and **underdispersion** if $\phi < 1$, which is far less common.
- The two most common quasi-likelihood models are:
  - the **quasi-Poisson** model, with $\mathrm{var}(Y_i) = \phi\,\mu_i$;
  - the **quasi-binomial** model, with $\mathrm{var}(Y_i) = \phi\,\mu_i(1-\mu_i)/m_i$.
- In both cases the estimating equations are the likelihood equations of the Poisson or binomial GLM, hence $\hat{\beta}$ is the **maximum likelihood estimate** of the GLM.

As noted in Unit C, the quasi-binomial model makes sense only for **grouped** data. With binary data $(m_i = 1)$ we necessarily have $\mathrm{var}(Y_i) = \mu_i(1-\mu_i)$, hence $\phi = 1$.

Where does overdispersion come from?

- **Unobserved heterogeneity**. Units with the same covariates may differ because of unrecorded variables. If $Y_i \mid U_i \sim \text{Poisson}(U_i\mu_i)$, with $\mathbb{E}(U_i) = 1$ and $\text{var}(U_i) = \sigma^2$, then

$$\mathbb{E}(Y_i) = \mu_i, \qquad \text{var}(Y_i) = \mathbb{E}\{\text{var}(Y_i \mid U_i)\} + \text{var}\{\mathbb{E}(Y_i \mid U_i)\} = \mu_i + \sigma^2\mu_i^2.$$

- **Correlated trials**. Let $S_i = m_iY_i = \sum_{j=1}^{m_i} Z_{ij}$, with $Z_{ij} \sim \text{Bernoulli}(\mu_i)$ having correlation $\rho > 0$, e.g. fetuses of the same litter. Then

$$\text{var}(Y_i) = \frac{\mu_i(1-\mu_i)}{m_i}\{1 + \rho(m_i - 1)\}.$$

- In both cases the inflation factor is **not constant**: variance inflation is only an **approximation**.

> Other variance functions are possible, e.g. $v(\mu) = \mu^2$ for counts, or $v(\mu) = \mu^2(1-\mu)^2$ for proportions (Wedderburn 1974), which corresponds to **no likelihood**. A different $v(\cdot)$ also changes the estimate.

The quasi-score

- The left-hand side of the estimating equations, divided by ϕ, is called **quasi-score**:

$$q_r(\beta) = \sum_{i=1}^{n} \frac{(Y_i - \mu_i)}{\text{var}(Y_i)} \frac{\partial \mu_i}{\partial \beta_r}, \qquad r = 1, \ldots, p.$$

- Under **(E.1)-(E.3)** alone, the quasi-score satisfies the **Bartlett identities** of a genuine score function, that is $\mathbb{E}\{q_r(\beta)\} = 0$ and

$$\mathbb{E}\{q_r(\beta) q_s(\beta)\} = -\mathbb{E}\left\{\frac{\partial q_r(\beta)}{\partial \beta_s}\right\} = \sum_{i=1}^{n} \frac{1}{\text{var}(Y_i)} \frac{\partial \mu_i}{\partial \beta_r} \frac{\partial \mu_i}{\partial \beta_s} = [\boldsymbol{X}^T \boldsymbol{W} \boldsymbol{X}]_{rs},$$

  where $\boldsymbol{W} = \text{diag}(w_1, \ldots, w_n)$ are the weights of Unit B.

> The quasi-score behaves like a genuine score, even though no distribution has been specified. This explains the name **quasi-likelihood**, and it is the key to the asymptotic results.

Properties of $\hat{\beta}$

Under the assumptions **(E.1)**-**(E.3)**, the estimator $\hat{\beta}$ is **consistent** and **asymptotically normal**:

$$\hat{\beta} \dot{\sim} \mathrm{N}_p\left(\beta, (\boldsymbol{X}^T \boldsymbol{W} \boldsymbol{X})^{-1}\right), \qquad w_i = \frac{1}{\mathrm{var}(Y_i) g'(\mu_i)^2} = \frac{1}{\phi} \frac{\omega_i}{v(\mu_i) g'(\mu_i)^2}.$$

- This is **formally identical** to the result of Unit B, but it now holds under much **weaker assumptions**.
- Does it make sense to talk about the **exact** distribution of $\hat{\beta}$? **No**: it would require the distribution of Y_i, which we deliberately left unspecified.
- Consistency only requires **(E.1)**: if the **mean** is correctly specified, $\hat{\beta}$ is consistent even when $v(\mu_i)$ is wrong. The **standard errors**, however, rely on **(E.2)**.

Estimation of ϕ and standard errors

- The dispersion ϕ is estimated with the **method of moments** estimator of Unit B, which is **consistent** under **(E.1)-(E.3)**:

$$\hat{\phi} = \frac{X^2}{n-p} = \frac{1}{n-p}\sum_{i=1}^{n} \omega_i \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}.$$

- The standard errors are obtained from $(\boldsymbol{X}^T \hat{\boldsymbol{W}} \boldsymbol{X})^{-1}$, plugging in $\hat{\beta}$ and $\hat{\phi}$. The weights w_i are **inversely proportional** to ϕ.

Variance inflation in a nutshell

When $v(\cdot)$ is the variance function of a GLM, as in the quasi-Poisson and quasi-binomial models:

1. Fit the **ordinary GLM** and keep its maximum likelihood estimate $\hat{\beta}$.
2. Multiply the **standard errors** of the GLM by $\sqrt{\hat{\phi}}$, that is $[\widehat{\text{se}}_{\text{quasi}}(\hat{\beta})]_j = \sqrt{\hat{\phi}}\,[\widehat{\text{se}}_{\text{glm}}(\hat{\beta})]_j$.

The quasi-log-likelihood

- To compare nested models we need something like a log-likelihood. Under **(E.1)-(E.3)** we define the **quasi-log-likelihood**

$$\ell_Q(\beta, \phi) = \sum_{i=1}^{n} \int_{y_i}^{\mu_i} \frac{\omega_i (y_i - t)}{\phi\, v(t)} \, \mathrm{d}t, \qquad \mu_i = g^{-1}(\boldsymbol{x}_i^T \beta).$$

- Its derivatives are the **quasi-score** $q_r(\beta)$. Indeed, if $F(\cdot)$ is a primitive of $t \mapsto \omega_i(y_i - t)/\{\phi\, v(t)\}$, then

$$\frac{\partial}{\partial \beta_r}\{F(\mu_i) - F(y_i)\} = F'(\mu_i)\frac{\partial \mu_i}{\partial \beta_r} = \frac{(y_i - \mu_i)}{\mathrm{var}(Y_i)}\frac{\partial \mu_i}{\partial \beta_r}.$$

- In general $\ell_Q(\beta, \phi)$ is **not** a log-likelihood. However, in the Poisson case ($\omega_i = 1$, $v(t) = t$, $\phi = 1$) we get $\ell_Q(\beta, 1) = \sum_{i=1}^{n}(y_i \log \mu_i - \mu_i)$, up to constants, which is the **Poisson log-likelihood**.

Comparing nested models

- The quantity $D(\boldsymbol{y};\boldsymbol{\mu}) = -2\phi\,\ell_Q(\beta,\phi)$ is called **quasi-deviance**. When $v(\cdot)$ is the variance function of a GLM, it coincides with the **deviance** of Unit B, e.g. the Poisson or the binomial one.
- Let $M_0 \subset M_1$ be two **nested models**, with p_0 and p parameters, and let $q = p - p_0$. The **quasi-log-likelihood ratio** statistic is

$$W = 2[\ell_Q(\hat\beta,\hat\phi) - \ell_Q(\hat\beta_0,\hat\phi)] = \frac{D(\boldsymbol{y};\hat{\boldsymbol{\mu}}_0) - D(\boldsymbol{y};\hat{\boldsymbol{\mu}})}{\hat\phi} \dot\sim \chi^2_q,$$

  where $\hat\phi$ is estimated from the **larger** model.

- The (quasi-)deviances must **not** be used as goodness-of-fit statistics, since ϕ is unknown.
- There is no likelihood, therefore the **AIC** is **not available**.

The Ants data

The quasi-Poisson model

- With the same mean structure, we obtain $\hat{\phi} = 4.18$: the standard errors are multiplied by $\sqrt{\hat{\phi}} = 2.04$.

	Estimate	SE (Poisson)	SE (quasi)	p-value (Poisson)	p-value (quasi)
(Intercept)	3.386	0.075	0.153	<0.001	<0.001
Bread2	0.154	0.061	0.124	0.011	0.221
Bread3	0.026	0.063	0.128	0.684	0.843
Bread4	-0.030	0.064	0.130	0.634	0.817
Filling2	0.097	0.088	0.180	0.272	0.594
Filling3	0.468	0.081	0.166	<0.001	0.008
Butter1	0.063	0.089	0.182	0.478	0.730
Filling2:Butter1	0.248	0.119	0.244	0.038	0.316
Filling3:Butter1	0.277	0.111	0.226	0.012	0.229

- The **estimates** are identical. After the correction, only `Filling3` remains **significant**.

Model simplification

- The larger standard errors suggest a **simpler** model. We remove one term at a time, using the quasi-log-likelihood ratio statistic W and its χ^2_q approximation.

Model	Removed term	q	W	p-value
Full model	Bread	3	2.581	0.461
Full model	Filling:Butter	2	1.621	0.445
Without Bread	Filling:Butter	2	1.614	0.446

- Neither `Bread` nor the `Filling`-`Butter` interaction is relevant, once overdispersion is accounted for.
- The final model only includes `Filling` and `Butter`, with $p = 4$ parameters.

The final model

	Estimate	Std. Error	p-value	exp(Estimate)
(Intercept)	3.318	0.103	$<$0.001	27.596
Filling2	0.233	0.120	0.059	1.262
Filling3	0.621	0.111	$<$0.001	1.860
Butter1	0.263	0.089	0.005	1.301

- Under the logarithmic link, $\exp(\beta_j)$ is the **multiplicative change** of the mean number of ants (Unit D). The estimated dispersion is $\hat{\phi} = 4.07$.
- Compared to Vegemite, **ham and pickles** attract on average about 86% more ants, and peanut butter about 26% more (borderline significant).
- The presence of **butter** increases the mean number of ants by about 30%. The type of bread is irrelevant.

Confidence and prediction intervals

- Consider a sandwich with ham and pickles and butter, whose estimated mean is $\hat{\mu}_{\text{new}} = 66.8$ ants. The 95% intervals are:

	Lower	Upper
Confidence interval for the mean	57.5	77.5
Prediction interval (quasi-Poisson)	34.5	99.1
Prediction interval (Poisson)	51.0	83.0

- The **confidence interval** for μ is obtained on the linear predictor scale and then transformed.
- Without a distribution, the **prediction interval** relies on the approximation $Y_{\text{new}} \dot{\sim} \text{N}(\hat{\mu}_{\text{new}}, \hat{\phi}\hat{\mu}_{\text{new}})$.
- The Poisson prediction interval, based on the Poisson quantiles, is **far too narrow**.

Summary

- A **quasi-likelihood model** only specifies the mean $g(\mu_i) = \boldsymbol{x}_i^T \beta$ and the variance $\mathrm{var}(Y_i) = (\phi/\omega_i) v(\mu_i)$ of independent responses.
- The estimate $\hat{\beta}$ solves the quasi-score equations via IRLS, and it is consistent and asymptotically normal under these weaker assumptions.
- The dispersion is estimated by $\hat{\phi} = X^2/(n-p)$. In the quasi-Poisson and quasi-binomial models, $\hat{\beta}$ is the GLM estimate and **all** standard errors are multiplied by $\sqrt{\hat{\phi}}$.
- Nested models are compared through the difference of deviances divided by $\hat{\phi}$. The AIC is not available.
- In practice, overdispersion is **very common** with counts and grouped proportions. Ignoring it leads to **overconfident** conclusions, as in the `Ants` data.

References

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Salvan, A., Sartori, N., and Pace, L. (2020), *Modelli lineari generalizzati*, Springer.

Wedderburn, R. W. M. (1974), "Quasi-likelihood functions, generalized linear models, and the gauss-newton method," *Biometrika*, 61, 439–447.